Dinesh Kumar Yadav

Backend Engineer | Data Scientist | AI Engineer
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Professional Summary

Backend Engineer, Data Scientist and AI Engineer (Computer Science undergraduate) experienced in **Machine Learning, Deep Learning and Generative AI** and in building production backends with **FastAPI and Node.js**. Built and deployed live products including a **RAG-based assistant** and an **online examination portal**, trained CNN models (**89% accuracy** on emotion recognition), and completed research internships in **Medical Image Segmentation and Quantum Machine Learning** at Symbiosis International University. Twice selected as **Team Lead** for institute-level Smart India Hackathon.

Technical Skills

Languages: Python, SQL, JavaScript, Java, C, C++
Machine Learning: Scikit-learn, Pandas, NumPy, Classification, Regression, Clustering, Feature Engineering, Cross-Validation
Deep Learning: TensorFlow, Keras, PyTorch, CNN, U-Net, Transfer Learning, Image Segmentation, Data Augmentation, OpenCV
Generative AI: LLMs, RAG, Prompt Engineering, Vector Embeddings, Semantic Search, Conversational AI
Quantum ML: Variational Quantum Circuits, Hybrid Quantum-Classical Models
Backend and Databases: FastAPI, Node.js, Express.js, REST APIs, JWT Authentication, MongoDB, MySQL, PostgreSQL
Frontend: React.js, Next.js, Tailwind CSS
Tools and Deployment: Docker, Git/GitHub, Postman, Jupyter, Google Colab
Evaluation: Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix, Dice, IoU

Research Experience

Research Intern – Quantum Machine Learning Aug 2026 – Present
Symbiosis International University (SIU) *Pune, India*
- Implementing **variational quantum circuits and hybrid quantum-classical models** for classification tasks and benchmarking them against classical baselines.
- Studying quantum data encoding techniques for mapping classical datasets into quantum feature spaces.

Research Intern – Medical Image Segmentation June 2026 – July 2026
Symbiosis International University (SIU) *Pune, India*
- Developed a **U-Net model** for automated Meibomian gland segmentation from infrared meibography images.
- Built preprocessing and augmentation pipelines and evaluated segmentation using **Dice Coefficient and IoU**.

Projects

MCAET College Information Assistant | *RAG, LLM, Python, FastAPI* | Live | GitHub 2026
- Built a **RAG conversational assistant** for college courses, admissions, departments and facilities using document chunking, embeddings and semantic search.
- Exposed the pipeline through a **REST API** and integrated it as a chatbot on the live college website.

MCAET Online Examination Portal | *React.js, FastAPI, MongoDB* | GitHub 2026
- Built a live exam system with secure **Student, Faculty and Admin dashboards** and authentication flows.
- Developed FastAPI REST APIs for exam scheduling, question handling and result processing across departments.

GlowMart India – E-Commerce Backend | *Node.js, Express, MongoDB, Python* | GitHub 2026
- Built REST APIs for catalog, cart, orders and **JWT authentication**, with optimized MongoDB schemas and product search.
- Implemented a recommendation module based on browsing and purchase patterns, exposed through a backend endpoint.

Facial Emotion Recognition | *CNN, TensorFlow/Keras, OpenCV* | GitHub Jan 2026
- Trained a CNN to classify **6 facial emotions** from 4,500+ grayscale images, achieving **89% accuracy and 0.87 weighted F1-score**.
- Applied data augmentation and dropout regularization and analyzed confusion-matrix based misclassification patterns.

Cattle Breed Classification | *CNN, Transfer Learning, Computer Vision* | GitHub 2025
- Developed a classifier for multiple Indian cattle breeds, achieving **80% accuracy**; used augmentation to handle an imbalanced image dataset.

Achievements & Leadership

- **Smart India Hackathon (SIH):** Cleared institute-level internal round twice (2025, 2026); served as **Team Lead** for architecture, backend and final presentation.
- Selected for two consecutive research internships at **Symbiosis International University** in applied AI domains.

Education

Acharya Narendra Deva University of Agriculture & Technology (ANDUAT) Ayodhya, UP
B.Tech, Computer Science and Engineering | ***CGPA: 7.5/10*** *Aug 2024 – May 2028*

Certifications

- Python for Data Science – Simplilearn
- AI Predict (Python) Workshop – SkillManthan Technologies Pvt. Ltd.
- Participant, AlgoUtsav 2026 Hackathon